

Local Labor Markets in Mexico:

definition, data bases, and descriptive analysis

1990-2020

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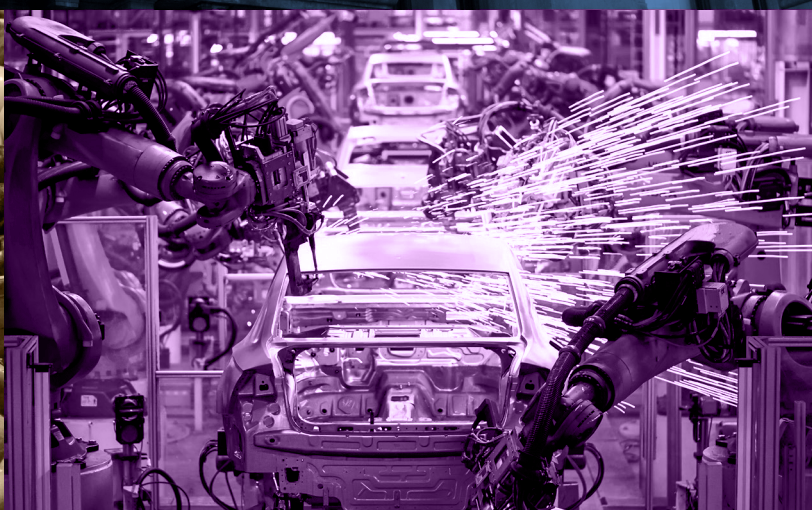
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2024



Banco de México's Data Lab (EconLab)



Local Labor Markets in Mexico: Key Insights

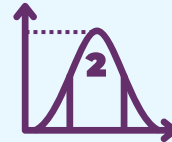
A local labor market is a set of municipalities that are linked between them, because the workers who live there work in the same set of municipalities. Local labor markets cover the entire country.



2,453 municipalities
in México in 2020



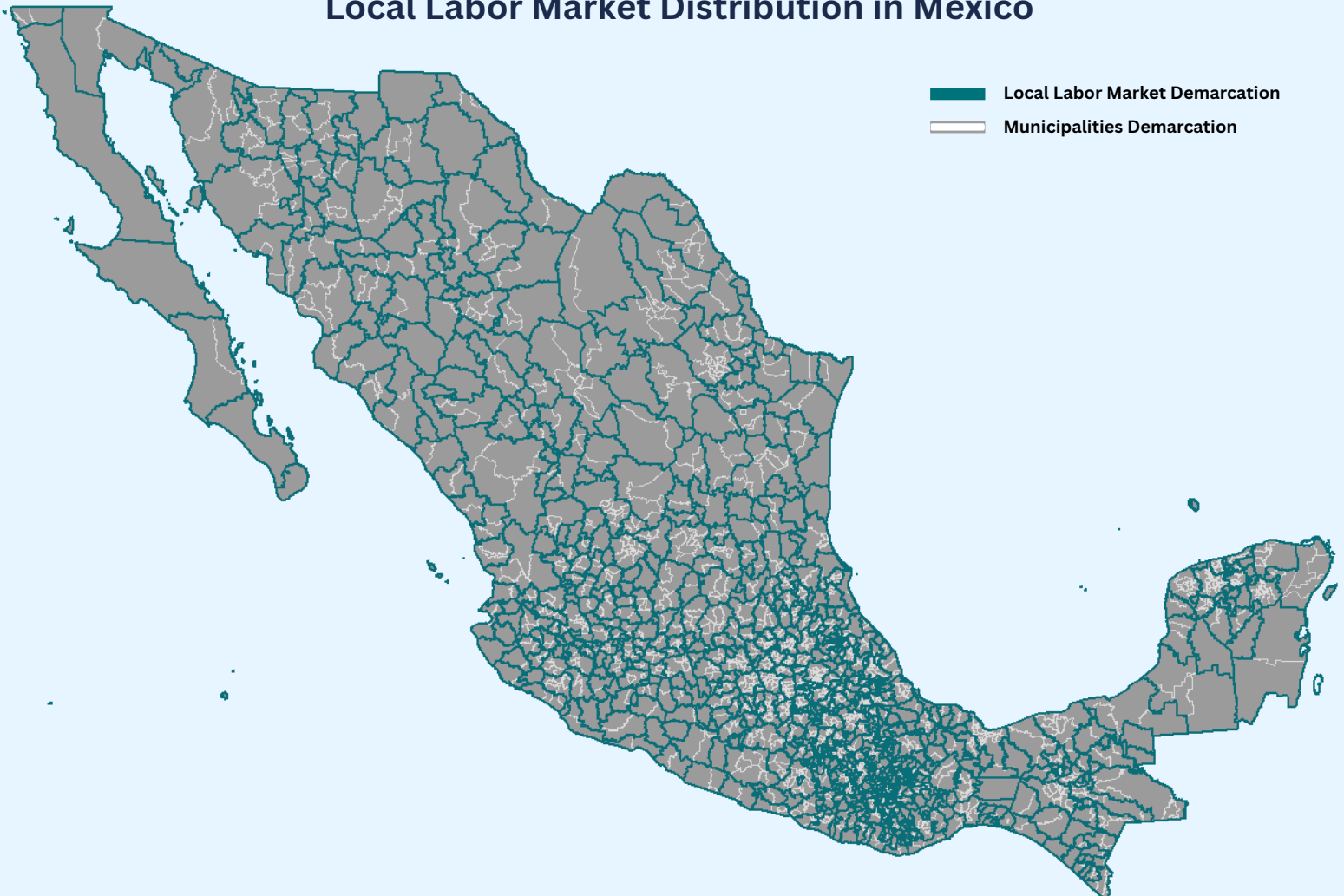
777 local labor
markets



The median local labor
market is made up of 2
municipalities

94.2%  of work trips are made within the same local
labor market

Local Labor Market Distribution in Mexico



Presentation: local labor markets in Mexico

To promote the study of Mexico's labor market, the Microdata Laboratory of Banco de México (EconLab) releases databases that include local labor market definitions for Mexico and relevant characteristics of the population and workers in these markets from 1990 to 2020. Econlab publishes these databases and their documentation for public use.

This document describes the methodology used to group all the municipalities in the country into 777 local labor markets. The approach used to define local labor markets follows best international practices and hinges on the idea that there are groups of municipalities that are highly economically integrated and, therefore, constitute a single local labor market. After defining local labor markets, census population and housing data collected by INEGI are linked to local labor markets, covering the period from 1990 to 2020. With the definitions generated in this document, researchers from the academic community and the public and private sectors can study different aspects of the Mexican labor market using a unit of analysis that considers the interrelationships between the country's municipalities.

The databases are presented at the individual level and aggregated at the local labor market level from 1990 to 2020. This document details the structure of the databases, and provides an initial analysis of the trends exhibited by the Mexican labor markets during 1990-2020 based on the aforementioned database.



EconLab's main objective is to promote the economic research carried out by this Central Institute through collaboration with national and foreign researchers. The EconLab of Banco de México makes the data mentioned in this document available to the public in the following link: [Databases \(SIDIE Datasets, Banco de México\) \(banxico.org.mx\)](https://banxico.org.mx/Databases/SIDIE-Datasets/Banco-de-México)

Any researcher interested in the use of microdata for economic research work can contact the EconLab at econlab@banxico.org.mx

Local labor markets are defined by grouping municipalities linked by transfer of workers



Source: Bank of Mexico

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1 Introduction

An important strand of the recent literature on labor markets has sought to identify the consequences of various shocks by exploiting heterogeneity in the exposure to them across local labor markets due to the lack of complete mobility of workers between locations (see, for example, Autor, Dorn and Hanson, 2013; Chiquiar, 2008 and Topalova, 2005 and 2010). Analyzing at the local labor market level allows for more detailed studies compared to using more aggregate information.

Any study of a country's labor market at a local level must first choose a unit of analysis that represents these local labor markets. Data on employment and wages are frequently reported by official sources at the geographic or political unit level, as is the case of municipalities or federal entities in Mexico. However, this level does not necessarily match the level at which economic relations occur or labor market variables are determined. For this reason, statistical agencies, academics, and international organizations have made efforts in several countries to generate more appropriate definitions of local labor markets from an analytical and economic point of view that serve as a basis for research on labor economics.¹

Given the tools commonly used in labor economics, when investigating labor markets it is desirable that the units of analysis cover the entire territory of the country in question and that they reflect the scale at which the prices and quantities of employment exchanged in a market are determined. The local labor market is a unit of analysis that largely satisfies these conditions. A *local labor market* is an area where workers and employers are highly interrelated due to their proximity in geographic space. Therefore, they are subject to similar conditions and respond similarly to labor demand and supply shocks. As Moretti (2011) points out, the level of employment and wages are determined locally; consequently, by grouping municipalities that share jobs and workers, the statistical units of analysis better approximate a definition of the local labor markets they wish to study (Fowler and Jensen, 2020). The main objective of this study is to generate a definition of local labor markets for the case of Mexico to achieve this improved approximation and deepen the study of Mexican labor markets.²

¹ For example, the US government's Economic Research Service (ERS) publishes and updates definitions of *commuting zones*, grouping counties that share a workforce; the OECD has developed definitions of metropolitan areas in its member countries; Baragwanath et al. (2021) have defined urban areas in India using satellite images; and Rozenfeld et al. (2011) cluster statistical units at the census block level in the USA according to a clustering algorithm to generate city definitions.

² While there are geographic or political definitions that divide the country into subunits, such as municipalities, states, or regions, these do not necessarily represent areas within which workers and employers are closely integrated. On the one hand, states may be too large to reflect local changes, such as the arrival of investment or changes in infrastructure in a specific locality. On the other hand, two adjacent municipalities can share a workforce and employment opportunities, so the municipality is not an ideal unit of analysis either. Some definitions, such as metropolitan areas defined by the INEGI, group municipalities into a single unit of analysis. However, these metropolitan areas apply only to highly populated urban centers, restricting the set of places available for study.

In this context, this document establishes the foundations for the analysis of local labor markets in Mexico through three contributions:

1. A definition of local labor markets for Mexico. These local labor markets group together municipalities that share a workforce and job opportunities, and they cover the entire territory of the country.
2. A standardization of Census data covering the period from 1990 to 2020, along with the 2015 intercensal count, to provide databases at the individual and local labor market levels for those years.³
3. An analysis of the trends in the national labor based on the proposed definition of local labor markets.

To define the local labor markets, first, a measure of the intensity of commuting between each pair of municipalities is constructed using data on the workers' municipalities of residence and work from the 2010 Census. The municipalities are then grouped following the same methodology used by the United States Economic Research Service (ERS). This US government agency produces official definitions of commuting zones, frequently used in work that studies the labor market in that country (see, for example, Autor, Dorn, and Hanson, 2013). The details of the process that generates the definition of local labor markets for Mexico are in Section 2 of this document. That section also provides diagnostic metrics of the proposed definition of local labor markets, comparing them with other existing definitions. This document complements Blyde et al. (2020), who also propose a definition of commuting zones for Mexico using a different methodology.

With these local labor market definitions at hand, panel databases were constructed (where the observation unit is the local labor market) for the period from 1990 to 2020, based on the Population Censuses and Housing from 1990, 2000, 2010 and 2020, as well as the 2015 Intercensal Survey.⁴ These databases at the local labor market level are made available to the public, as well as the individual-level microdata on which the aggregates are based. To allow comparison between data from different years, the variables and catalogs of the microdata from the extended questionnaire of the Population and Housing Censuses and the Intercensal Count were standardized. The variables included in the databases at the labor market level are standardized throughout the study period and cover various topics related to the labor market, including i) demographic composition, ii) employment, iii) labor income measures; iv) social and economic vulnerability, and v) informality. Bartik shock variables were also constructed to approximate labor demand shocks, thus allowing the estimation of parameters related to the effects of changes in employment at the local level, such as the elasticity of labor supply or the relationship between employment and poverty.⁵ Section 3 of this document describes the structure of the bases and their construction. The databases, catalogs, codes, and other associated documentation can be found in the following link: [Base de datos \(SIDIE Datasets, Banco de México\) \(banxico.org.mx\)](https://banxico.org.mx/SIDIE_Datasets/Base_de_datos_(SIDIE_Datasets,_Banco_de_México)_).

Last, Section 4 of this document conducts an initial descriptive study on local labor markets in Mexico. This study seeks to illustrate how the published databases can be used and to establish some stylized facts about the Mexican labor market that can serve as a starting point for future research. Employment and wage trends are

³ The 2015 intercensal survey is included, but not those from 2005 or 1995, because the latter provide less detailed information about the economic characteristics of the respondents and, therefore, are not appropriate for studying employment and wage levels.

⁴ Data for the 2020 Census was collected between March 2 and March 27, 2020, before the Mexican Federal Government declared the start of the health emergency in the country on March 31st. Therefore, these data are not suitable to study the effects of the COVID-19 pandemic.

⁵ See Bartik (1991), Blanchard & Katz (1992), Goldsmith Pinkham *et al.* (2020), and Borusyak *et al.* (2022).



shown by schooling levels from 1990 to 2020, by region, to understand the conditions of labor supply and demand broadly. Indicators of inequality, poverty and informality are also shown.

2 Building local labor markets for Mexico

Any economic study has the selection of a unit of analysis as its starting point. As mentioned, in labor economics it has become increasingly common to define local labor markets as groups of geographic units that share a labor force and job opportunities. Fowler and Jensen (2020) point out some advantages of having local labor market definitions; in particular, they are appropriate for understanding regional variation in wages, differences in education and unemployment between regions, and the dynamics of business formation. Consequently, numerous studies of the labor market in the United States and other countries follow delimitations of this type (e.g., Autor et al., 2013; Autor et al., 2021; Card et al., 2018; Manning and Petrongolo, 2017; Monte et al., 2018). Specifically, many of the studies mentioned use the definition of local labor markets (*commuting zones*) published by the ERS.⁶ Its methodology, which is well-known and replicable, consists of grouping in the same local labor market the political units (in the case of the US, counties) between which workers move when traveling from their homes to their workplaces.

The ERS methodology is replicable for the case of Mexico using INEGI's data. While there are some other definitions of geographic units of analysis for the study of the labor market in Mexico, the local labor markets proposed in this paper may present some advantages to researchers over those alternatives. In Mexico, INEGI has created delimitations of metropolitan areas in 2000, 2005 and 2010. Although this delimitation is helpful for city analysis and urban planning, it only encompasses a fraction of the territory in Mexico. This lack of coverage does not allow an analysis of smaller local labor markets, which are not necessarily close to a highly populated city, and therefore exclude a significant part of the national labor market. Blyde et al. (2020) propose a delimitation of commuting zones based on expanding existing metropolitan zones and grouping municipalities not associated with these existing zones into new ones. This delimitation allows the entire Mexican territory to be covered. However, it poses two limitations. First, it considers each rural municipality not associated with a metropolitan area as a single commuting zone, without considering possible links between rural municipalities close to each other. This criterion generates many small rural commuting zones in their approach. Second, their methodology differs from the United States' definition of commuting zones. Maintaining comparability to the United States is convenient for studying the effects of policies previously studied in the United States, such as the effect of imports from China (Autor et al., 2013) or local minimum wages (Allegretto et al., 2017). Since these analyses use definitions of local labor markets based on the *ERS methodology*, it is possible to replicate them through the local labor markets presented here and thus compare the economies of the United States and Mexico.

2.1 Construction of local labor markets for Mexico

To have a delimitation of local labor markets that covers the entire national territory, integrates small and interrelated rural areas, and is methodologically comparable with the existing measures in the United States, this document defines local labor markets using the methodology from Fowler and Jensen (2020). This methodology was used to construct the commuting zones for 2010 in the United States and follows that of Tolbert and Sizer (1996), which in turn was used to construct commuting zones for 2000. The central idea is that if two geographical units are in the same single labor market, they should share the same workforce. Therefore, workers who live in a municipality within the local labor market tend to work within the same market, although possibly in another municipality. On the other hand, a reduced flow of commuters would be observed between two municipalities that belong to different local labor markets. Following Fowler and Jensen (2020), *local labor*

⁶ See <https://www.ers.usda.gov/data-products/commuting-zones-and-labor-market-areas/>

markets are “observations where the high density of commuting relationships among subunits suggests that changes in the supply of or demand for labor in one subregion would be felt equally within other subunits within the region” (Fowler and Jensen, 2020).

In the case of the United States, the geographic sub-units usually grouped into local labor markets are counties or census blocks. As mentioned above, data about commuting to work in Mexico is available at the municipal level, so municipalities are used as a basic sub-unit. Starting from the municipal limits defined in the national geostatistical framework of the 2010 Population and Housing Census,⁷ the municipalities are associated using a clustering algorithm and data on trips to work between municipalities. First, for each pair of municipalities indexed by i, j a dissimilarity index D_{ij} is calculated. This index is smaller when those two municipalities are more closely integrated in terms of the flows of workers between them:

$$D_{ij} = 1 - \frac{f_{ij} + f_{ji}}{\min(\sum_l f_{il}, \sum_l f_{jl})}. \quad (1)$$

In this equation:

- The variable f_{ij} is the number of people who live in municipality i and work in municipality j ;
- The variable f_{ji} is the number of people who live in municipality j and work in municipality i .
- Thus, D_{ij} is calculated as one minus the sum of the number of commuters between both municipalities, divided by the workforce of the municipality that has the smallest number of residents. In the denominator, the index l represents each of the country's municipalities in the sum. Hence, the term $\sum_l f_{il}$ is the total number of residents of the municipality i .

To exemplify the calculation of the dissimilarity index D_{ij} , suppose there are two municipalities, i and j , such that all the workers in both municipalities work in j . In that case, if i has a population of workers greater than j , the numerator of the second term will be equal to the denominator, and the dissimilarity index will be zero, indicating that the two municipalities are (entirely) integrated in terms of their workforce. In general, when the number of travelers between municipalities i and j is high relative to the labor force, the second term of the equation tends to be larger and the dissimilarity index between municipalities tends to be small, indicating a high level of integration of the two municipalities.⁸

Once the dissimilarity indices are defined, the country's municipalities are grouped according to this index using a hierarchical grouping algorithm (*clustering*). The algorithm begins by considering all municipalities in a single local labor market. Subsequently, it divides this labor market into two to maximize the average dissimilarities between the two groups. Then, it repeats this process until 777 local labor markets are obtained. This number of markets is the same number of commuting zones defined by the exercise carried out by Blyde et al. (2020).⁹

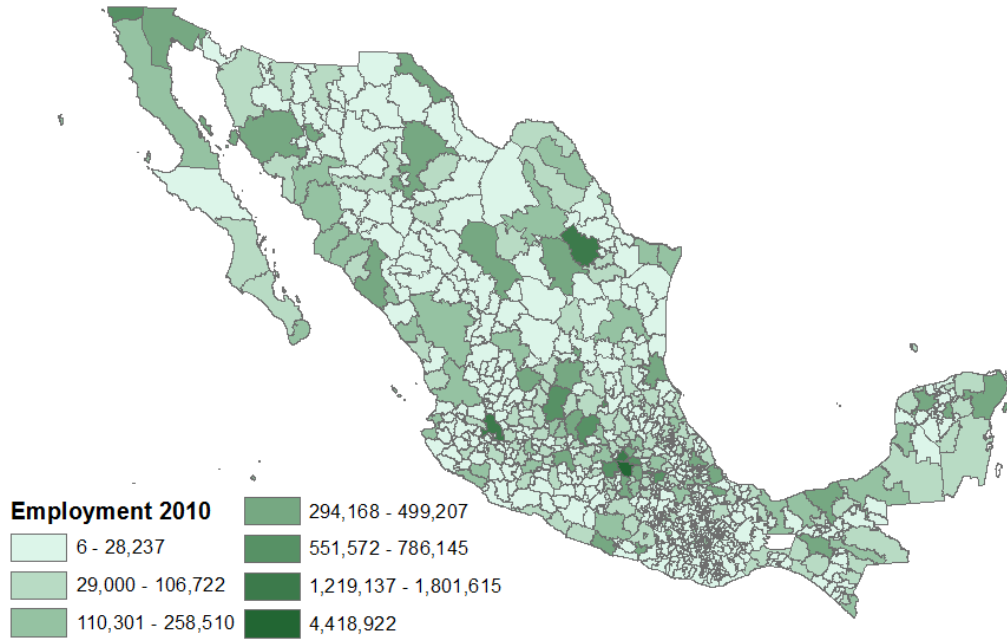
⁷ National Institute of Statistics and Geography (2010). *Geostatistical framework 2010 version 5.0A*, 2010 Population and Housing Census, Mexico. Available at <https://www.inegi.org.mx/app/biblioteca/ficha.html?upc=702825292805>.

⁸ In cases where the index is negative, its value is replaced by zero. In addition, municipalities that are more than 150 kilometers apart are defined to have a dissimilarity index of one.

⁹ An alternative approach is to set a threshold for the average dissimilarity between groups, above which groups should separate. Tolbert and Sizer (1996) applied this methodology to the United States, setting the dissimilarity threshold at 0.98. Using this threshold in Mexico results in 925 commuting zones. This number of zones exceeds the 777 Blyde et al. (2020) found, and many resulting zones are small and rural. For this reason, it was decided to set the number of commuting zones rather than using a dissimilarity threshold to select the groups.

Figure 1 shows a map of the 777 local labor markets found with our approach, according to their level of employment in 2010. Mexico City (CDMX), along with other proximal municipalities, is contained into a single commuting zone. The average employment in the commuting zones identified with this algorithm in 2010 was 54.6 thousand people.

Figure 1: Map of local labor markets



Source: Prepared by Banco de México with information from INEGI.

2.2 Descriptive statistics of commuting zones and diagnostic measures

The local labor markets generated, along with some diagnostic measures, are described below. The first column of Table 1 shows descriptive statistics on the constructed local labor markets. The median number of municipalities in a local labor market is two municipalities, with a maximum of 30 municipalities in a single local labor market. The set of local labor markets defined here covers all of the country's employment in 2010. The largest local labor market, located in Mexico City's metropolitan area, has approximately 4.5 million workers. Column 2 shows the same statistics for the commuting zones constructed by Blyde et al. (2020). These are typically smaller, with a median of one municipality per commuting zone. However, the largest commuting zone according to the methodology of Blyde et al. (2020), Mexico City, is bigger than the largest local labor market according to the methodology proposed here.

Table 1 shows some diagnostic measures of local labor market connectivity and compares them with the same measures for commuting zones from Blyde et al. (2020). These connectivity diagnoses suggested by Fowler and Jensen (2020) are measures of the degree to which the labor market study units are self-contained, i.e., to what degree workers tend to remain in the same market on their trips from home to work. In the average local labor market identified in this study, 94.2% of work trips are made within the local labor market. This measure contrasts with 83.9% for an average commuting zone in Blyde et al. (2020). The local labor markets defined here are also less likely to divide the metropolitan areas produced by INEGI into different local labor markets.

Table 1: Descriptive statistics of local labor markets and comparison with those obtained with the methodology of Blyde et al. (2020)

Local labor markets	Current (Tolbert and Sizer Methodology, 1996)	Methodology of Blyde et al. (2020)
Number of local labor markets	777	777
Median number of municipalities in each local labor market	2	1
Maximum number of municipalities in a local labor market	30	74
Maximum number of workers in a local labor market	4,418,922	7,153,860
INEGI 2015 metropolitan areas divided into more than one local labor market ¹⁰	11	16
Proportion of total trips to work that are made within the local labor market	0.942	0.954
Proportion of trips to work that occur within the average local labor market	0.904	0.839
Minimum proportion of trips to work that occur within a local labor market	0.290	0.251

Source: Prepared by Banco de México with information from INEGI and Blyde et al (2020).

¹⁰ According to INEGI, there were 74 metropolitan areas in Mexico in 2015.

3 Databases at individual, housing-unit, and local labor market levels

A central objective of producing definitions of local labor markets is to facilitate the generation of high-quality research on the Mexican labor market. With this intention, this document provides researchers with data that offers flexibility and convenience for these analyses at the local labor market level. For this purpose, three databases (at the individual, housing unit, and local labor market levels) from the Population and Housing Censuses from 1990 to 2020 and from the 2015 Intercensal Survey are made available to the public on a website dedicated to this purpose.¹¹ The values for the variables in these databases are standardized between years, and include identifiers of the local labor market to which each observation belongs to under the definition generated in this document.¹² The structure of the databases, their content, and the variables standardization process are described in the remainder of this section.

The first two databases (at the individual and housing unit level) are derived from the microdata provided by INEGI in the Population and Housing Censuses and are standardized across years. With these data, researchers can generate alternative variables they might require in their specific analysis. The database of individuals includes demographic, work-related, and educational variables. The second data set contains housing-unit microdata that can be linked to microdata belonging to individuals through the identifiers included in both databases. This second database includes variables on the physical characteristics of the dwelling, the services available in it, and the household that inhabits it. Finally, aggregated databases are provided at the local labor market level, with demographic, employment, informality, social vulnerability variables, and Bartik shocks, a variable that approximates local labor demand shocks and is common in labor economics studies. All the micro-level bases include the geographic identifiers provided by INEGI for each data item, along with a variable that indicates the local labor market to which the observation belongs. The rest of this section describes how the individual and household-level data were extracted and standardized and the resulting databases. Subsections detail some specific considerations for the databases of individuals, housing, and local labor markets.

One of the challenges in comparing local labor markets in the study period (1990-2020) is that the census and intercensal survey questionnaires that provide data on labor markets have changed over the years. Therefore, to provide comparable data, the codebooks of different waves were compared, the compatible variables were selected, and the catalogs that describe their values were standardized. The objective was to preserve as many variables as possible from each census and maintain comparability between years. Information about the variables and their homologation process is included in the methodological note accompanying this document and can also be found on the dedicated website.

For categorical variables for which INEGI provides a catalog of valid values, a “Not specified” value was assigned as part of the classification so that these variables never have missing entries. For categorical variables for which values cannot be calculated with INEGI-sourced data, the value “Not specified” was also assigned. Variables with a numerical value are assigned the quantity of the variable they represent, and their units are detailed in the documentation. Unlike categorical variables, if there is no value, these numerical variables may be empty.

¹¹ The webpage is available at [Bases de datos \(SIDIE Datasets, Banco de México\) \(banxico.org.mx\)](https://bases.de.datos(SIDIE%20Datasets,%20Banco%20de%20M%C3%A9xico)(banxico.org.mx)).

¹² Although INEGI conducted Population and Housing Censuses in 1995 and 2005, these censuses did not provide as detailed information on employment and wage variables as the 2015 Intercensal Survey. Therefore, they are unsuitable for analyzing local labor markets during those periods. Consequently, the 1995 and 2005 Censuses are not included in this project.



Data sets were harmonized using a star model, where the data is in a central file with numerical references to all catalogs.¹³ For database record at the individual level, a unique identifier was constructed with the period and the individual's identifier. In the same way, the bases at the housing or household level include variables that identify these units. All bases include a variable that identifies the local labor market to which they belong, constructed according to the description in Section 2.

For variables for which each observation could have a set of several values, it was necessary to use additional tables integrated into the star model. This is the case for variables such as the individual's different sources of income, access to different healthcare systems, and disabilities. In these files, the unique identifier of each record is kept, and as many entries as necessary for each case are added.¹⁴

The individual and housing-unit-level data include INEGI's geographic identifier variables to combine these datasets with others. Further, variables that identify economic sectors are coded using the first three digits of the North American Industrial Classification System classification (NAICS).

3.1 Census microdata (individual level)

This dataset has all the individual variables from the extended questionnaires from the 1990, 2000, 2010, and 2020 Population Censuses and the 2015 Intercensal Count. The 1990 source data combines the individual and household data, so the variables from the original bases were classified into these two categories to include them in the corresponding database. The variables' values, categories and structure were standardized across years.¹⁵ The general structure for the database at the individual level is described in a Data Dictionary, which accompanies this document and is on the website mentioned above. The Methodological Note details some considerations of the standardization process, such as the construction of person identifiers and the harmonization of input variables.¹⁶

3.2 Census microdata (housing-unit level)

Like the database at the individual level, this information was extracted from the extended census questionnaires, but in this case, it was done using the housing-unit level databases. For 1990, there was no division between both sets of variables in the source Census data, so the source variables were sorted into the individual or housing-unit level databases according to their description. The details of the standardized values and the criteria considered for each year are in the Methodological Note. The variables are described in the corresponding Data Dictionary.¹⁷

¹³ Under this model, the central data file has only numerical information without text, which minimizes the occurrence of errors when loading the file with statistical processing software. See Kimball and Ross (2013) for details on the flagship model.

¹⁴ For example, for the disability variables, if an individual has a disability, their record will appear in this additional file indicating the type of disability. If the individual has more than one disability, it reappears as many times as necessary. Individuals without disabilities do not appear in these additional files.

¹⁵ The details of the standardized values and the criteria taken into account for each year can be consulted in the Methodological Note on the databases on the dedicated webpage, [Databases \(SIDIE Datasets, Banco de México\) \(banxico.org.mx\)](https://banxico.org.mx/SIDIE_Datasets).

¹⁶ The highest quality survey available to measure income in Mexican households is the National Household Income and Expenditure Survey (ENIGH), carried out every two years and representative at the federal entity level. While labor income data from censuses may be less precise, they are representative for each municipality in the country and, therefore, are reported and leveraged here.

¹⁷ The details of valid values for the housing database can be found in the corresponding Data Dictionary, in the link [Databases \(SIDIE Datasets, Banco de México\) \(banxico.org.mx\)](https://banxico.org.mx/SIDIE_Datasets).

Unlike later censuses, for 1990 and 2000 the housing-unit-level source data include variables that describe the households that live in a specific unit and may vary within a housing unit. For both years, separate household and housing identifying variables were constructed.

3.3 Databases at the local labor market level

To study the local labor markets of Mexico, and as one of the central contributions of this project, a panel database was built at the local labor market level that covers the period from 1990 to 2020, using as a basis the definitions of local labor markets described in Section 2. The database provides variables to study local employment and wages, as well as average demographic and educational characteristics that can be determining factors for labor supply and demand. Given the high prevalence of informality in the country, the database also includes estimates of the percentage of people in this type of employment. All these variables are measured separately for the total population and men and women. Measures of economic and social vulnerability are also available, including poverty and extreme poverty rates, child labor, single-parent households, and the population that does not study or work.¹⁸

Finally, labor demand shock variables are provided (based on Bartik, 1991, and Goldsmith-Pinkham et al., 2020), which, together with the definition of local labor markets, allow the estimation of various economic relationships associated with the labor market, such as the elasticity of labor supply.¹⁹ Specifically, the labor demand shock variables are defined as follows:

$$B_{lt}^G \equiv \sum_k s_{lk,1990} g_{kt}^G. \quad (2)$$

In this equation, l is the local labor market, and t represents the year. The demand shock B_{lt}^G for all workers ($G = \text{Todos, All}$), men ($G = \text{Hombres, Males}$) or women ($G = \text{Mujeres, Females}$) is a weighted average of employment growth in each industry k at the national level between the previous period and year (t) for gender (G), g_{kt}^G , which is calculated excluding the market local labor l .²⁰ The weights $s_{lk,1990}$ are the shares of industry employment k in total employment in the local market l in 1990.

The databases at the local labor market level are listed in the Data Dictionary corresponding to the aggregated databases. The Stata codes that generate the aggregate databases from the individual-level microdata are also available on the web page.²¹ Finally, correspondences from municipalities to local labor markets are provided for users to generate their own variables at the local labor market level. Thus, it is possible to produce the aggregates that the user requires at the local labor market level from the microdata of persons and households.

¹⁸ Although there is housing data at the municipal level that could be used to generate housing information at the local labor market level, the definitions of those aggregate variables are reserved for a future stage of this project. Likewise, incorporating variables based on INEGI's migration information modules are within project's scope in the next stage.

¹⁹ A recent application of these shocks can be seen in Banco de México (2023).

²⁰ The periods included are 1990, 2000, 2010, 2015, and 2020. Because data for two consecutive periods are needed to calculate the terms g_{kt}^G only data on demand shocks from 2000 and afterward are included.

²¹ The Stata codes are found under the "Códigos para generar agregados" section, in the [Databases link \(SIDIE Datasets, Banco de México\) \(banxico.org.mx\)](https://banxico.org.mx/SIDIE_Datasets).



4 Local labor markets in Mexico between 1990-2020

This section describes the Mexican labor market from 1990 to 2020 using the standardized data and databases at the local labor market level described in the previous section. The section provides some stylized facts about how local labor markets changed throughout the study period, as a starting point for future work.

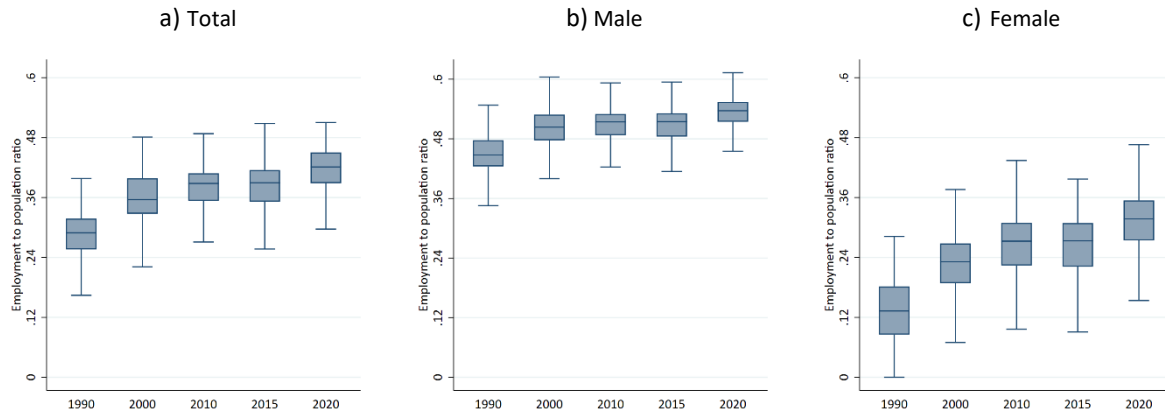
The observation unit in the following analysis is the local labor market, constructed as described in Section 2 unless otherwise noted. Employment and wages are analyzed by schooling level, region, and gender. These three dimensions of heterogeneity allow empirical observations to be related to structural changes or shocks that vary in their incidence, such as the opening to international trade observed throughout the 1990s, as well as other more recent shocks (the effects of entry from China to the WTO, the global crisis of 2009, and the trade war between the United States and China, among others). In the same way, this level of analysis allows a first approximation to the evolution of inequality and regional income in the country.

Employment is classified into two groups according to their level of education in order to consider the differences in human capital between workers: skilled workers are assumed to correspond to individuals with 16 or more years of schooling, while unskilled workers are those with less than 16 years. Two common indicators in the labor economics literature are calculated to study the differences between these two groups (Autor et al., 2008): the ratio of the wages of skilled to unskilled workers (the *skill premium*) and the ratio of skilled employment to unskilled employment (the *skill ratio*). These concepts allow for the evaluation of changes in the relative supply and demand of skill, that is, of skilled with respect to unskilled labor.

4.1 Employment and wages

Figure 1a shows the employment to population ratio, which increased between 1990 and 2020. Employment growth was relatively greater for women: between 1990 and 2020, the percentage of working women went from approximately 13% to more than 30%. In this context of increases in employment, the variation over time in wages would allow us to infer whether the expansion was mainly due to labor supply or demand increases, which can also be studied by skill level. Table 2 shows that, among unskilled workers, from 1990 to 2000, real wages fell significantly, by 20.3%. Although in this decade, the Mexican labor market experienced the crisis of 1994-1995, the increase in employment and the decrease in real wages suggest that an expansion in labor supply may have been observed during the period. This expansion is noticeable especially in the case of female employment.²² The wage reduction in this period was substantial enough so that real unskilled wages in 2020 remain slightly below the 1990 level. From 1990 to 2000, a decrease in real wages was also observed among workers with higher levels of education, although this decrease was of a smaller magnitude in percentage terms.

²² Various studies have documented the notable increase in female labor participation in Mexico. See, for example, Calderón (2014) and López Acevedo et al. (2021).

Figure 1: Employment to population ratio, 1990-2020

Note: Excludes outliers. The outliers are all those that are below or above the “whiskers”, which are estimated as follows: Upper whisker: $p_{75} - 1.5 \cdot IQR$; Lower whisker: $p_{25} + 1.5 \cdot IQR$. Here, IQR is the interquartile range defined as the difference between the third (p_{75}) and first quartile (p_{25}).

Source: Prepared by Banco de México with information from INEGI.

During the decade from 1990 to 2000, the North American Free Trade Agreement (NAFTA) came into force, which reduced trade barriers between Mexico, the United States, and Canada. Due to its proximity to the border, Mexico’s northern region exhibited more substantial impacts from the treaty, such as a relatively higher increase of labor demand, especially for unskilled workers (Chiquiar, 2008, and Chiquiar et al., 2017). Indeed, unskilled workers’ wages decreased less in the northern region in the 1990s than in other regions. Between 2000 and 2010, real wage increases were observed in both skill groups, suggesting that the demand for labor expanded beyond possible increases in labor supply in this decade. Finally, between 2010 and 2020, wages diminished again, indicating that labor supply may have grown faster than demand for both skill groups.

Table 2: Unskilled and skilled wages, 1990-2020**a) Average wage (1990 National Average Wage Index = 100)**

Year	National		North		North-Central		Central		South	
	Level	% Change	Level	% Change	Level	% Change	Level	% Change	Level	% Change
1990	100.00		118.18		100.49		104.49		78.02	
2000	89.41	-10.59	113.34	-4.09	88.49	-11.94	91.84	-12.10	67.92	-12.94
2010	107.50	20.23	122.70	8.26	108.01	22.06	109.76	19.51	91.58	34.84
2020	104.58	-2.72	125.19	2.03	108.80	0.73	104.57	-4.73	83.80	-8.49

b) Unskilled wage (1990 National Average Wage Index = 100)

Year	National		North		North-Central		Central		South	
	Level	% Change	Level	% Change	Level	% Change	Level	% Change	Level	% Change
1990	94.25		109.26		95.51		95.63		77.88	
2000	75.07	-20.34	97.28	-10.96	75.57	-20.88	75.78	-20.75	57.00	-26.81
2010	89.72	19.50	104.68	7.61	91.76	21.42	90.45	19.35	75.67	32.75
2020	86.82	-3.23	106.87	2.09	92.17	0.46	84.99	-6.03	69.14	-8.62

c) Skilled wage (1990 National Average Wage Index = 100)

Year	National		North		North-Central		Central		South	
	Level	% Change	Level	% Change	Level	% Change	Level	% Change	Level	% Change
1990	218.92		242.69		197.49		234.02		173.29	
2000	206.98	-5.46	235.68	-2.89	193.43	-2.06	215.51	-7.91	173.63	0.20
2010	208.09	0.54	227.37	-3.52	199.81	3.30	214.76	-0.35	185.87	7.05
2020	181.28	-12.88	207.04	-8.94	177.79	-11.02	186.16	-13.32	151.83	-18.31

d) Skill premium

Year	National		North		North-Central		Central		South	
	Level	% Change	Level	% Change	Level	% Change	Level	% Change	Level	% Change
1990	2.32		2.22		2.07		2.45		2.23	
2000	2.76	0.43	2.42	0.20	2.56	0.49	2.84	0.40	3.05	0.82
2010	2.32	-0.44	2.17	-0.25	2.18	-0.38	2.37	-0.47	2.46	-0.59
2020	2.09	-0.23	1.94	-0.23	1.93	-0.25	2.19	-0.18	2.20	-0.26

Note: % Change refers to the percentage change between periods.

Source: Prepared by Banco de México with information from INEGI.

Table 3: Skilled employment, 1990-2020

a) Ratio of skilled working-age to total working-age population										
Year	National		North		North-Central		Central		South	
	Level	% Change	Level	% Change	Level	% Change	Level	% Change	Level	% Change
1990	0.05		0.05		0.04		0.06		0.03	
2000	0.07	0.03	0.08	0.03	0.07	0.03	0.08	0.03	0.06	0.03
2010	0.11	0.04	0.12	0.04	0.11	0.04	0.12	0.04	0.09	0.04
2020	0.15	0.04	0.15	0.03	0.15	0.05	0.16	0.04	0.13	0.04

b) Ratio of skilled employment to total employment										
Year	National		North		North-Central		Central		South	
	Level	% Change	Level	% Change	Level	% Change	Level	% Change	Level	% Change
1990	0.07		0.08		0.07		0.09		0.05	
2000	0.10	0.03	0.11	0.03	0.10	0.03	0.11	0.03	0.08	0.03
2010	0.15	0.05	0.16	0.05	0.15	0.05	0.16	0.05	0.13	0.05
2020	0.19	0.04	0.19	0.04	0.20	0.05	0.20	0.04	0.17	0.04

c) Skill ratio (ratio of skilled employment to unskilled employment)										
Year	National		North		North-Central		Central		South	
	Level	% Change	Level	% Change	Level	% Change	Level	% Change	Level	% Change
1990	0.09		0.10		0.09		0.11		0.07	
2000	0.12	0.03	0.13	0.03	0.12	0.03	0.13	0.03	0.10	0.03
2010	0.18	0.06	0.19	0.06	0.18	0.06	0.19	0.06	0.16	0.06
2020	0.24	0.06	0.24	0.05	0.25	0.07	0.25	0.06	0.21	0.05

Note: % Change refers to the absolute variation between periods.

Source: Prepared by Banco de México with information from INEGI.

Regarding the labor market by skill level, we observe that the ratio of skilled to unskilled employment increased from 1990 to 2020 in all regions. This fact reflects an increase in the average schooling of the population: skilled employment increased as a proportion of the working-age population throughout the study period, as seen in Table 3.²³

Given that the ratio of skilled to unskilled employment increases throughout the period, the skill premium allows us to infer whether the relative demand or supply of skill increased more between each successive pair of observed years, as shown by Tables 2 and 3. From 1990 to 2000, the ratio of skilled to unskilled wages increased, suggesting that the relative demand for skilled labor grew more than the relative supply of skilled workers during this period. Notably, the increase in the skill premium was smaller in the northern region of the country, which may be associated with the increase in the demand for relatively less qualified labor in that region after the entry into force of NAFTA.²⁴

²³ Education at various levels became mandatory in the mentioned years. In particular, compulsory secondary school came into effect in 1993, and in 2009, three years of compulsory preschool were established. Likewise, throughout these years, the Progres-Oportunidades-Prospera program was implemented, which contributed to an increase in schooling among the poorest households in the country (see, for example, Parker 2019 for a general discussion of the effects of this program on education).

²⁴ Chiquiar (2008) presents evidence in this sense.



In the decades after 2000, the results suggest a progressive growth in the relative supply of skilled labor since, as Tables 2 and 3 show, an increase in the skilled employment ratio was observed, accompanied by a decrease in the skill premium between 2000 and 2010, and again between 2010 and 2020. Esquivel (2011), Campos-Vázquez (2013), and Levy and López Calva (2019) discuss the factors behind the reduction in returns to skill in the period from 1990 to 2010. The data shown in Table 1 indicates that this trend persisted during the decade from 2010 to 2020. The northern region experienced smaller declines in the skill premium relative to other regions in those two periods, suggesting that the demand for skilled labor increased more in this region in those years.

4.2 Wages and schooling

The evolution of wages and schooling is described below.²⁵ Figure 2 shows the changes in real wages by years of schooling in each region by decade studied, comparing them with the national changes. The vertical axis shows logarithmic differences in wages by educational level, averaged across local labor markets in each region.

From 1990 to 2000, real wages for individuals with more years of schooling increased slightly, while they decreased for those with fewer years of schooling. The positive slope observed in the log wage difference between 1990 and 2000 confirms that the skill premium increased during that decade. On the other hand, starting in 2000, the growth pattern is the opposite: the negative slope indicates that the skill premium decreased in both decades, as mentioned in the previous subsection. From 2000 to 2010, real wages increased on average for workers with less than or equal to 12 years of schooling, while they decreased slightly for those with a higher educational level. In the subsequent decade, from 2010 to 2020, a reduction in the skill premium was also observed, reflected in the negative slope of the line of changes in the logarithm of national wages between 2010 and 2020. This change is observed in the context of a reduction in real wages for most levels of schooling, which suggests an expansion in labor supply at all levels of education.

From a regional perspective, the north behaves slightly differently from the national average throughout the three decades, possibly because of its proximity to the United States and the associated greater exposure to external shocks. From 1990 to 2000, the reduction in real wages for workers at the lowest levels of schooling was smaller in the north than in other regions. This smaller reduction may be due to the positive demand shock for relatively less skilled labor associated with the entry into force of NAFTA. In the following decade, between 2000 and 2010, the wage increase for the lowest levels of schooling is lower in the north than in other regions. One possible cause for the less favorable wage increases of workers with low education in the north is that the region was more exposed to the 2008 crisis, which originated in the United States. In the last decade, on the other hand, salaries showed larger changes in the north than in the rest of the country, suggesting that the region experienced a positive demand shock with an impact on all levels of education.

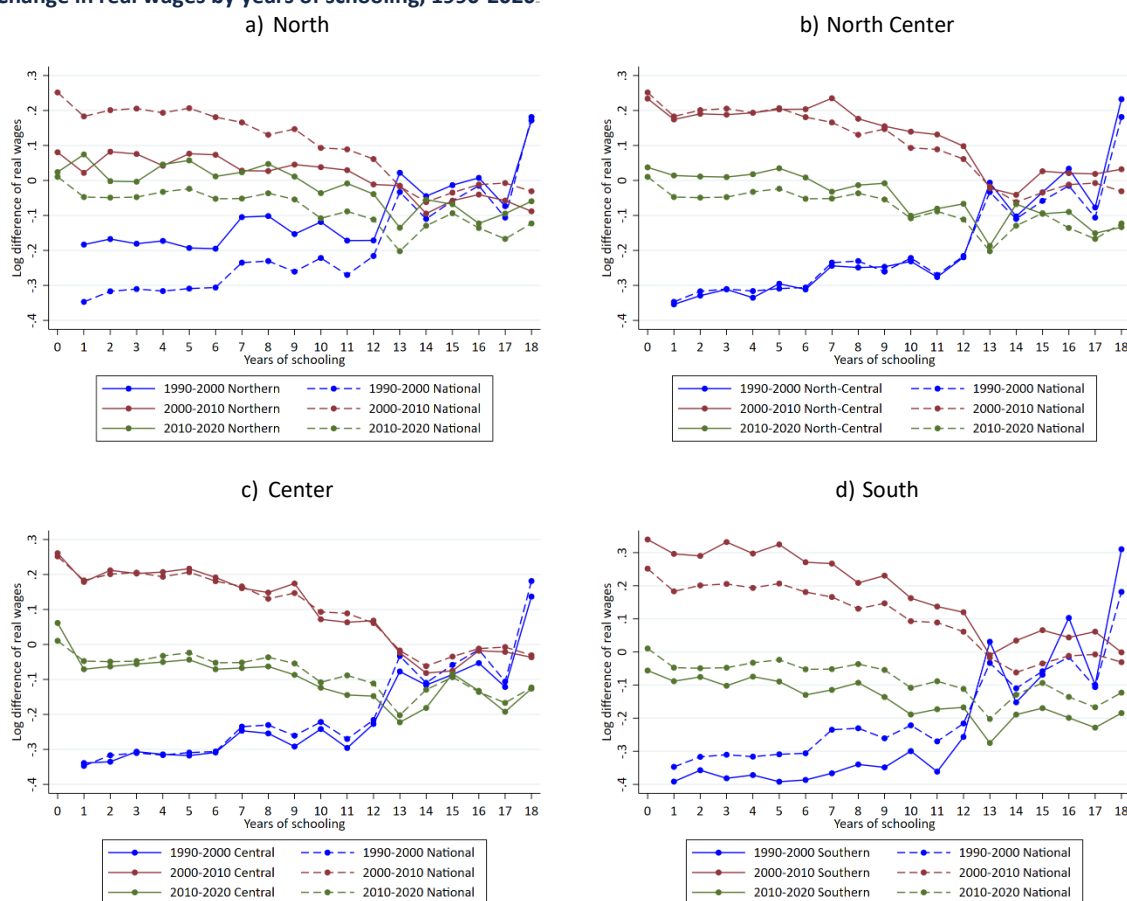
In the southern region, wages tend to vary in the same direction as the national average but with more pronounced changes. In the first period studied, from 1990 to 2000, the wage reduction mentioned above was greater in the south than in the national average, especially for the lowest levels of schooling. This greater reduction may reflect an especially strong increase in labor supply. The following period, from 2000 to 2010, was characterized at the national level by an increase in real wages for all educational groups, with wages in the south

²⁵ Regarding the estimation of returns to education, see Harberger and Guillermo-Peón (2012) and Caamal (2017). These estimates can be made using individual data such as those made publicly available with this document.

increasing even more. Due to the lower initial level of wages in this region, this implied regional convergence, as shown below. After 2010, the southern region again showed larger wage reductions at all educational levels.

The central and north-central regions tended to exhibit wage changes that were very similar to each other and to national patterns. However, between 2010 and 2020, wages for the individuals with the lowest levels of education in the north-central region exhibited a more favorable evolution than in the rest of the country. This evolution may also be associated with the greater linkage of that region with international markets.

Figure 2: Change in real wages by years of schooling, 1990-2020²⁶



Source: Prepared by Banco de México with information from INEGI

4.3 Residual wages and wage convergence

Table 1 shows that wages vary significantly between the regions of the country: in 1990, the average wage of unskilled workers in the southern region was equivalent to 77.9% of the national average salary, while in the north, the salaries of the same group were on average 109.3% of national salaries. Dispersion in wages contributes fundamentally to explain the income inequality of a country. For this reason, it is relevant to understand the sources of differences in returns to labor between skill groups and regions. The literature on wage dispersion in the United States has considered three central factors to explain inequality in labor income: differences in workforce composition, especially concerning education and experience; differences in returns to

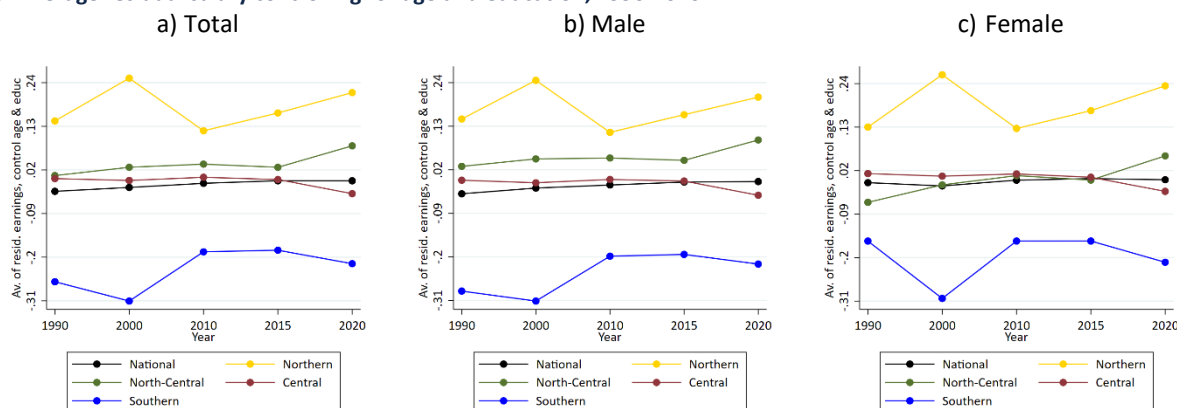
²⁶ For the 1990 Census, the category zero years of schooling is not included, but is grouped in the “Not specified” group.

schooling; and residual variation unrelated to the educational or demographic characteristics of workers (see as an introduction to this literature, for example, Autor and Katz, 1999; Autor et al., 2008, and Lemieux, 2006). This field of research has found that, for the United States, workers' schooling levels and experience of workers and the returns associated with these characteristics explain only one-third of the observed wage dispersion. For this reason, the dispersion in the so-called "residual wage," that is, the part of wages that does not depend on the educational or demographic characteristics of the workers, explains most of the growing inequality in the country. To assess this same issue in the case of Mexico, the residual wages of Mexican workers are estimated below, and their dispersion is examined throughout the three decades.

To calculate residual wages, separate linear regressions are estimated for each year, using worker-level data. The dependent variable is the logarithm of wages, and the independent variables are different observable categories, such as the educational level and age.²⁷ The regression residuals are named residual wages and are compared between regions and between men and women for each year in the sample (including 2015) in Figure 3.

Residual wages are higher in the northern region for all periods and for both men and women, indicating that they contribute to regional inequality in wages. This result suggests that labor demand, institutions, or unobservable productivity-related characteristics of workers may be greater in the north, implying higher returns to work even for individuals with the same observable characteristics. The opposite phenomenon is observed in the southern region, where returns to work are consistently lower, even after controlling for workers' education and age.

Figure 3: Average residual salary controlling for age and education, 1990-2020



Source: Prepared by Banco de México with information from INEGI.

Residual wages change over the years and between regions, reflecting changes in returns to work, and inducing adjustments in the distribution of labor income within the country. Between 1990 and 2000, a regional divergence pattern appears due to the increase in residual wages in the north and a reduction in the south, which is consistent with the positive shock of labor demand in the country's northern region as a result of NAFTA. In the subsequent decade, the opposite pattern is observed, with a reduction in returns to labor in the north and a corresponding increase in the south, which together led to a convergence in the labor income of the country's

²⁷ Age serves as a proxy variable for experience. Fourteen five-year age groups are used, ranging from 15 to 79 years, plus those over 80. For its part, with respect to the education variable, four categories are used: 1) Maximum six years (basic education); 2) between 7 and 12 (average); c) between 13 and 15 years old (higher average); and d) 16 or more (higher). In one regression, the independent variable is the categorical variable age, while in the second, it is the interaction between both categorical variables.

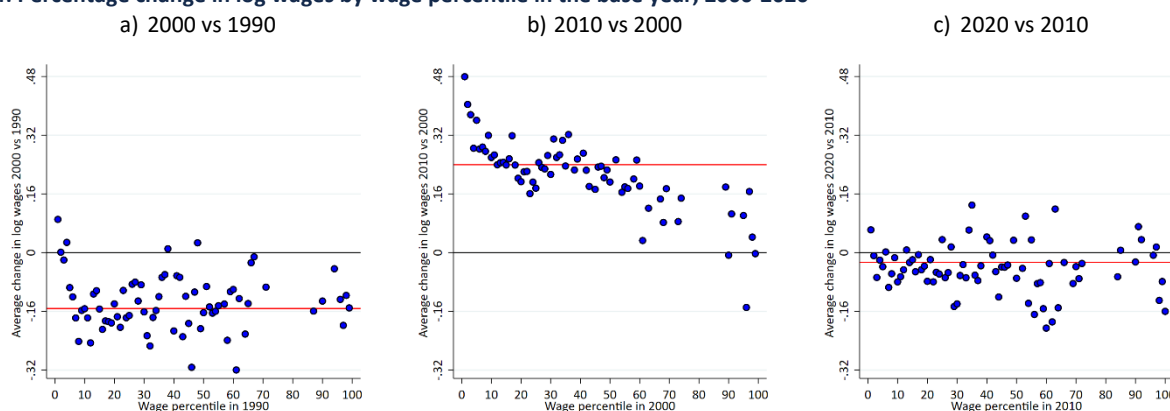
workers and a reduction in inequality, as shown in Section 4.4. below.²⁸ Over the following years, from 2010 onwards, a gradual divergence pattern in residual wages can be noted, with reductions in wages in the center and south and increases in those in the center-north and north. In general, the observed patterns are similar between male and female workers.

4.4 Wage convergence between local labor markets

The data constructed in this document allows an analysis of the distribution of wages in the country's local labor markets and their changes over time. Thus, it also allows for the analysis of labor income inequality between different local markets. As a first illustrative exercise on changes in the geographical distribution of wages, Figure 4 shows the percentage changes in wages in local labor markets in each decade studied, ordered by the wage level in the base year.²⁹ Between 1990 and 2000, wage growth rates were not strongly correlated with the initial level of wages in local labor markets. However, from 2000 to 2010, there was a clear negative relationship between wages in 2000 and the growth rate of local labor market wages, indicating geographical convergence. Wages in the average local labor market increased by approximately 24% in this decade and at higher rates where starting wages tended to be initially lower. This result indicates that the geographical convergence mentioned in the previous section was generally reflected within the country. Between 2010 and 2020, however, wage growth was again weakly linked to baseline levels.

Figure 5 shows different measures of wage inequality between the country's labor markets. To complement the changes observed in different parts of the distribution, wage ratios across local labor markets are shown at the 90th, 50th, and 10th percentiles and the standard deviation of the entire distribution.

Figure 4: Percentage change in log wages by wage percentile in the base year, 2000-2020



Note: Each point represents a percentile in the distribution of wages across local labor markets in the base year. The red lines represent the median of the distribution of changes in log average salary in the period.

Source: Prepared by Banco de México with information from INEGI.

The ratio between wages at the 90th and 10th percentiles is a measure of inequality that depends more on the behavior of the tails of the wage distribution. In 1990, the local labor market in the 90th percentile had a wage

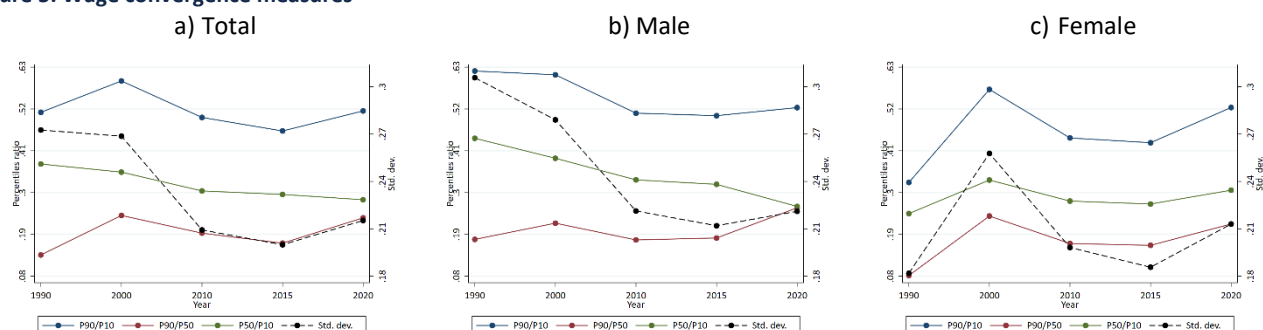
²⁸ It is possible that the shock associated with the 2008 crisis has affected workers in the north more strongly, while the rise in oil prices, which could have increased the productivity of that sector in the south, has contributed to regional convergence.

²⁹ The graphical representation of the percentage growth in income for each percentile is known as the Growth Incidence Curve (GIC). A curve with a negative slope would imply a redistribution that favors the first percentiles. For more information, see Peragine et al. (2014) and the World Bank's LAC Equity Lab data platform in the economic growth section: <https://www.worldbank.org/en/topic/poverty/lac-equity-lab1/overview>.

approximately 52% higher than the market in the 10th decile. This ratio increased around 2000, reflecting that the decade's growth resulted in a slightly more unequal wage distribution. Consistent with previous observations, after 2000, a significant compression was observed in the geographical distribution of salaries towards 2010 and 2015, which by 2020 began to reverse. However, when analyzing the wage ratio between the 50th percentile and the 10th percentile, we observe that the inequality between these two points of the salary distribution decreased between each pair of years analyzed, from which it may be concluded that the increase in inequality between the 90th and 10th percentiles appears to be primarily due to the separation between the labor market median and those in the upper tail of the distribution.

Figure 5 also shows measures of wage inequality between local labor markets by gender. It is notorious that with the substantial entry of women into the labor force after 1990, under all the measures shown, there was a marked increase in inequality in average female earnings between local labor markets. Inequality in women's salaries appears to increase more between 2015 and 2020 than in male salaries.

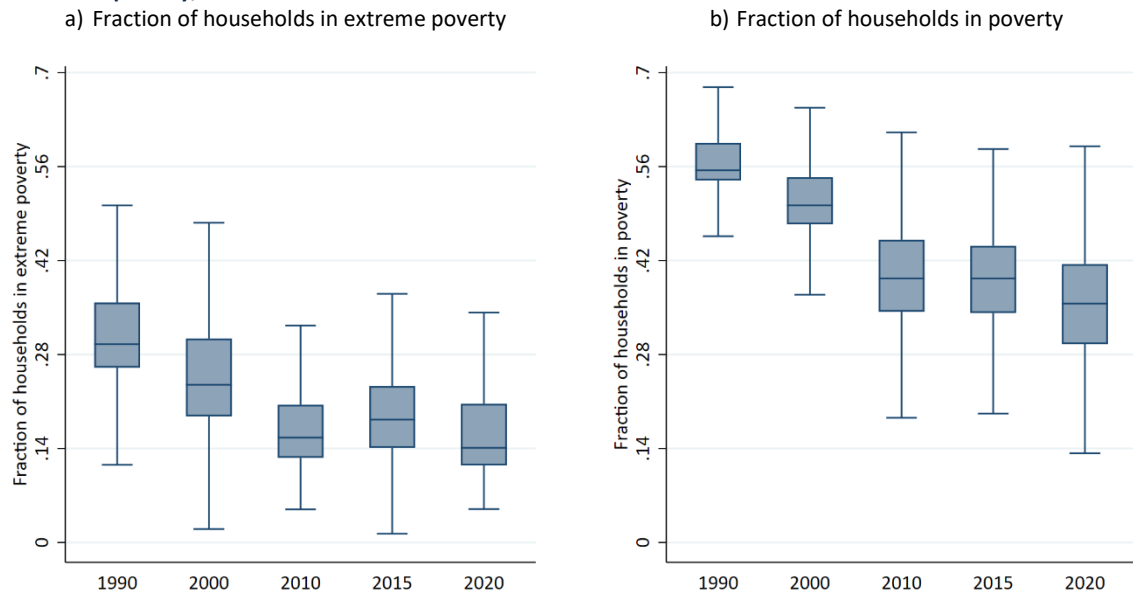
Figure 5. Wage convergence measures



Source: Prepared by Banco de México with information from INEGI.

Finally, the evolution of poverty rates at the household level is shown throughout the study period, following the official definitions of poverty and extreme poverty. Between 1990 and 2010, both poverty measures decreased significantly: in the median local labor market, the percentage of households in extreme poverty decreased from approximately 29% to approximately 16%. For its part, poverty went from 56% to 30%. Although these changes occurred over two decades, and at what appears to be a similar rate between each two years, the previous analysis suggests that the reduction in poverty rates observed between 1990 and 2000 has had different causes from that observed between 2000 and 2010. The first period was characterized by a significant entry of workers into the labor market, especially women and unskilled workers.³⁰ On the other hand, given that the decade from 2000 to 2010 was characterized by an increase in the wages of less skilled workers, it is possible that the observed poverty reduction in that period was due not to greater participation but to growth in real wages among lower-income workers.

³⁰ This suggests that the stronger incorporation of these workers into the labor market could have increased the income of households below the poverty line, for example, with the entry of women from housework into the labor market.

Figure 6. Household poverty, 1990-2020

Note: The poverty and extreme poverty lines published by CONEVAL are used, adjusting for changes in the price level between years.
 Source: Prepared by Banco de México with information from INEGI and CONEVAL.



5 Final remarks

This document presents a methodology consistent with the best international practices to generate definitions of local labor markets in Mexico. Local labor markets are built by grouping municipalities among which important flows of work-related trips are observed. Likewise, individual and aggregate-level databases are made available to the public to study these local labor markets. The purpose of this effort is to invite researchers from various sciences to carry out in-depth analyses of the evolution of labor markets in Mexico and thus enrich the research discussions within the country. To exemplify the usefulness of the databases and make some basic observations about the Mexican labor market from 1990 to 2020, a descriptive analysis of local labor market dynamics has been shown. This document should serve as a starting point for new research developments related to the Mexican economy and its labor markets.

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